



ASTROVITAL AI : VITALX CORE V1

CAREER & VISION DOCUMENT

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*"We did not wait for permission. We
did not wait for an institution. We
identified a twelve-year failure in
global aerospace medicine — and we
built the solution ourselves." —
Gourgopal Mohapatra, 2026*

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SECTION A — IDENTITY & VISION

Who Built This & Why:

A.i — The Researcher

The individual behind ASTROVITAL AI and the VITALX CORE V1 system is Gourgopal Mohapatra — a second-year undergraduate student pursuing a Bachelor of Technology in Artificial Intelligence and Data Science at an institution in India. This document does not originate from a funded laboratory, a government research body, or a corporate R&D division. It originates from the desk of a solo researcher who identified a critical, unresolved problem at the frontier of human spaceflight — and chose to solve it independently, without waiting for institutional authorization, without a research supervisor, and without external funding of any kind.

Every line of code, every trained model, every dataset record, every notebook, and every word of documentation in the VITALX CORE V1 project was produced by one individual operating on personal initiative and personal time. This is stated not as a footnote but as a defining characteristic of the work — because the independence of this research is itself a demonstration of the kind of initiative, technical depth, and mission-driven commitment that the aerospace industry demands of those who aspire to operate at its frontier.

A.ii — The Vision

The career trajectory that this work is designed to advance is not conventional. The immediate target is the Indian private space sector — a rapidly maturing ecosystem of deep-technology startups operating at the intersection of satellite systems, launch vehicle engineering, space domain awareness, and aerospace AI.

Within this landscape, ASTROVITAL AI is positioned as a flagship independent research project that demonstrates technical capability, research rigor, and a long-term commitment to the specific domain of aerospace health intelligence.



The longer-term trajectory points toward the global aerospace industry — toward organizations such as Airbus and Boeing, whose India R&D operations represent the most demanding and consequential application environments for the kind of systems developed under the ASTROVITAL program. The vision is not to contribute a single project to a single employer. The vision is to establish ASTROVITAL AI as a research and development program that grows in parallel with the researcher's career — version by version, capability by capability — until it constitutes a recognized, mission-deployable standard for autonomous crew health monitoring in human spaceflight. That program begins here. It begins with VITALX CORE V1. And it is already underway.

SECTION B — THE SPACE HEALTH CRISIS

WHY ASTRONAUTS NEED AI

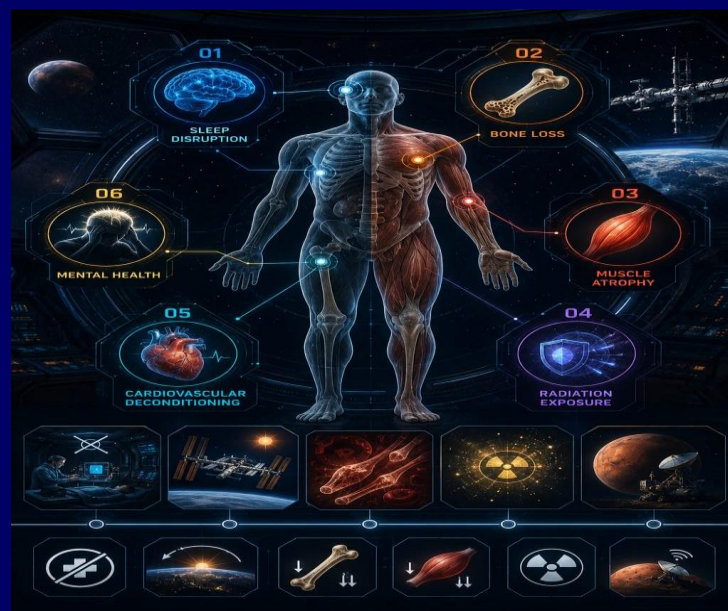
B.i — The Hostile Environment

Space is not merely an extreme environment — it is a systematically hostile one, engineered by physics to undermine human physiology at every level simultaneously. The International Space Station, orbiting at approximately 400 kilometers altitude, completes one full revolution of Earth every 90 minutes — exposing crew members to sixteen sunrises and sixteen sunsets within every 24-hour period. The circadian disruption this imposes is not a comfort issue; it is a clinical one. Sleep architecture degrades, hormonal regulation is disrupted, cognitive performance declines, and immune function is compromised — all before the crew has been exposed to a single one of the more acute threats that deep space missions introduce.

In microgravity, the human skeleton loses bone mineral density at a rate of approximately 1 to 2 percent per month — a rate that, over the duration of a Mars transit mission, would render bones structurally compromised without aggressive countermeasure protocols. Skeletal muscle mass degrades at rates of up to 16 percent over short-duration missions, with the postural and load-bearing muscle groups — precisely those most critical for emergency physical response — disproportionately affected. The cardiovascular system undergoes fluid redistribution toward the upper body, reshaping cardiac geometry, impairing orthostatic tolerance, and reducing the body's capacity to respond to hemorrhagic or pressure emergencies.

B.ii — The Radiation Reality

Radiation exposure in deep space represents one of the most irreducible physiological risks of human spaceflight. Within the Earth's magnetosphere, crew members aboard the International Space Station receive approximately 50 to 100 millisieverts of ionizing radiation per six-month mission — already ten to twenty times the annual occupational limit for radiation workers on Earth. Beyond the magnetosphere, on a trajectory to Mars, crew members are exposed to the full, unattenuated spectrum of galactic cosmic rays and solar energetic particles, with cumulative doses that current shielding technologies cannot reduce to acceptable levels.



The physiological consequences are not limited to elevated cancer risk, though that risk is significant. Acute radiation syndrome, central nervous system effects including cognitive degradation and motor impairment, immune suppression, and accelerated cardiovascular disease are all documented consequences of the radiation environment that deep space crews will inhabit. These effects do not announce themselves with clear, sudden symptoms — they accumulate silently, manifest gradually, and are most dangerous precisely because their early warning signals are subtle and easily attributed to other causes. A monitoring system that cannot detect these signals autonomously, continuously, and

without Earth support is not a monitoring system — it is a recording device.

B.iii — The Communication Barrier

The most operationally critical parameter of the deep space medical environment is not physiological — it is electromagnetic. A one-way radio transmission from Earth to a spacecraft in Mars transit takes approximately 24 minutes. A round-trip communication — a query sent from a crew medical officer, received by Earth-based physicians, assessed, and a response transmitted — takes a minimum of 48 minutes, and in practice considerably longer when accounting for processing and response formulation time. In any acute medical emergency, this latency is not a delay — it is a complete absence of support.

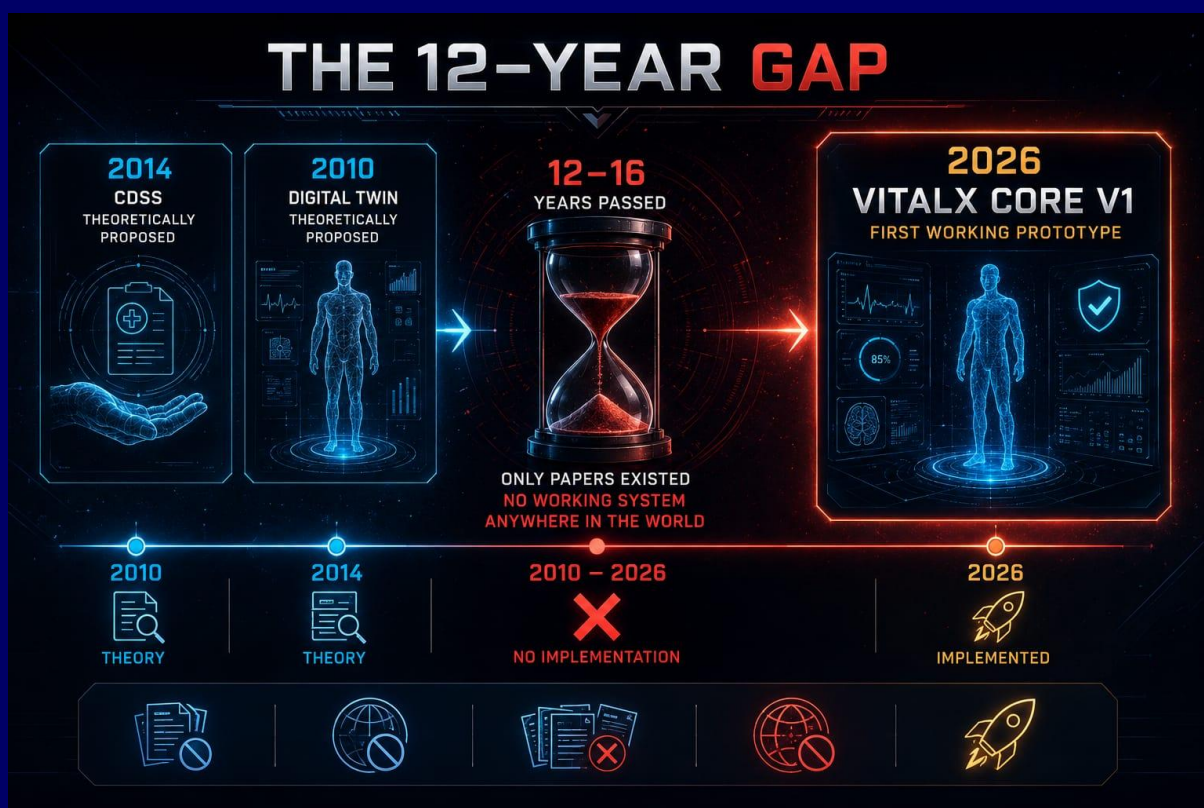
Cardiac events, acute neurological episodes, severe radiation syndrome, and hypoxic crises can reach irreversible clinical thresholds within minutes. No Earth-based physician, regardless of their expertise, can provide meaningful real-time guidance across a 24-minute communications gap. The crew is alone. Their health monitoring system must therefore be capable of autonomous, real-time, explainable clinical decision-making — not as an enhancement to Earth-based support, but as a complete and independent replacement for it. This is the operational reality that motivated the development of VITALX CORE V1, and it is the operational reality against which every design decision in the system must be evaluated.

SECTION C — THE 12-YEAR GAP

WHAT NOBODY BUILT

C.i — The Papers That Gathered Dust:

The scientific community recognized the need for autonomous clinical decision support in human spaceflight long before any functional system was built to address it. A Clinical Decision Support System for space medicine was formally proposed in peer-reviewed literature as early as 2014. The concept of a Digital Twin — a continuously updated computational model of an individual astronaut's physiological state — was proposed as early as 2010. These were not speculative ideas proposed by fringe researchers; they were documented, peer-reviewed, formally published recommendations produced by scientists working within the established space medicine research community.



What followed was sixteen years of papers citing other papers. Reviews citing reviews. Proposals citing previous proposals. The

research literature expanded. The implementation gap did not close. As of the beginning of 2026, no functional, edge-deployed, explainable clinical decision support system existed anywhere in the world — not in any NASA laboratory, not in any ESA research program, not in any university research group, and not in any private aerospace organization. The gap between the documented need and the delivered solution had persisted, unaddressed, for over a decade.

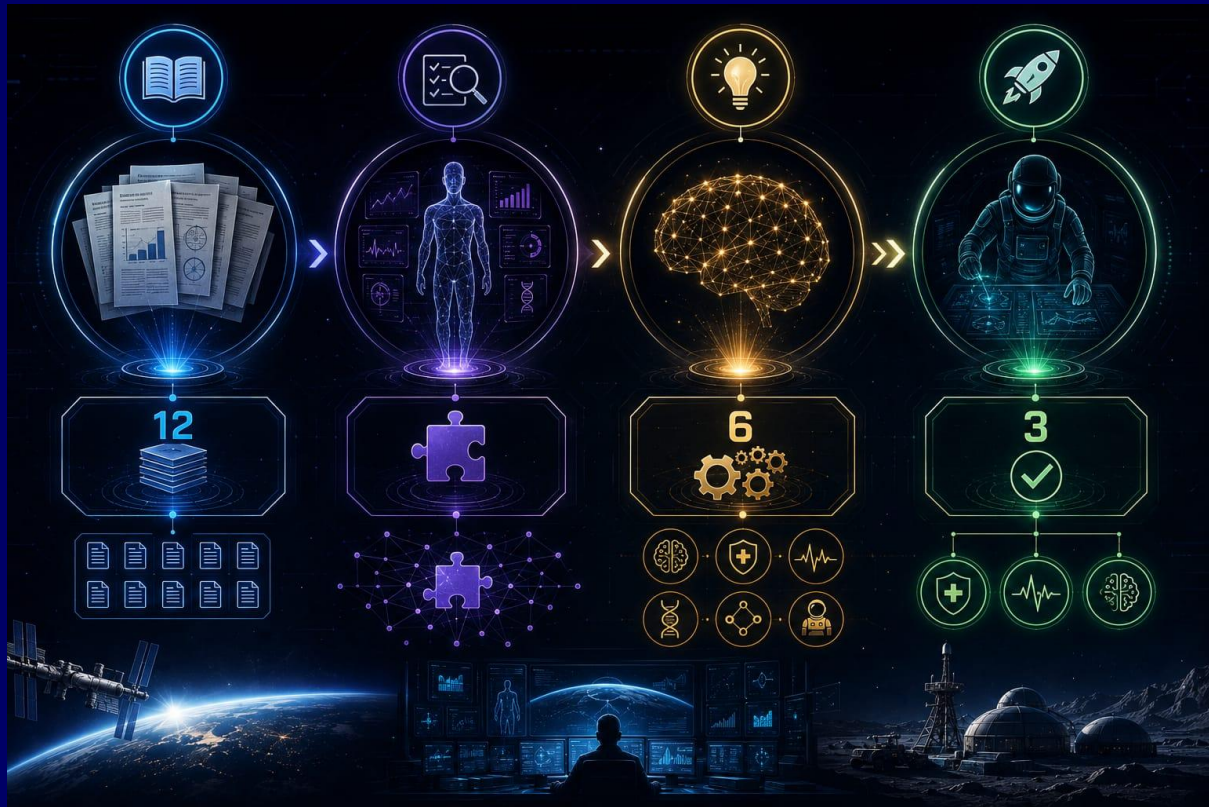
C.ii — Why This Matters:

The twelve-year implementation gap is not merely an academic curiosity — it is a mission risk. NASA's Artemis program is actively planning crewed lunar surface operations. Crewed Mars missions are on the strategic roadmaps of both national space agencies and private launch providers. These missions will carry human beings into environments where the medical risks are real, the communication barriers are absolute, and the consequence of an undetected health crisis is irreversible. The absence of a functional autonomous CDSS is not a research gap — it is an operational gap in the readiness of human spaceflight for the missions that are already being planned.

VITALX CORE V1, completed in May 2026, represents the first known functional prototype of a working, edge-deployable, explainable Clinical Decision Support System designed for human spaceflight physiological monitoring. It was not built by NASA. It was not built by ESA. It was not built by a funded university research program. It was built by a single second-year undergraduate student, working independently, in India — because the need was real, the gap was documented, and no one else had built the solution. That fact is significant. It will not be forgotten when the history of this program is written.

SECTION D — RESEARCH FOUNDATION

BUILT ON SCIENCE



D.i — The Literature Base

VITALX CORE V1 was not built on intuition, on assumption, or on generic machine learning tutorials. It was built on science. The research foundation of the system comprises twelve peer-reviewed papers spanning the full spectrum of spaceflight physiology: cardiovascular deconditioning, respiratory and hypoxic response, radiation-induced biological effects, musculoskeletal deterioration, neurological and cognitive performance, and psychological health in isolated and confined environments. Each of these papers was read in full, analyzed in detail, and subjected to a structured critical review focused not on what the research had achieved but on what it had explicitly failed to achieve.

This approach — reading not for knowledge but for gaps — is the methodological distinction that separates literature review as a research foundation from literature review as a formality. The goal was not to summarize what was known. The goal was to identify precisely what was missing, what had been proposed but never built, what had been recognized as a limitation but never addressed. Three weeks of dedicated study in space medicine, supplemented by this structured gap analysis across all twelve papers, produced the innovation architecture of VITALX CORE V1.

D.ii — From Gaps to Innovations

The gap analysis yielded six distinct and documentable deficiencies in the existing research and implementation landscape of space medicine AI. These six gaps defined six potential innovations — six specific technical problems whose solutions would constitute genuine contributions to the field rather than incremental refinements of existing approaches. VITALX CORE V1 implements three of these six innovations: Sensor Drift Correction, Biomarker Discovery, and the Edge Clinical Decision Support System. The remaining three innovations are allocated to future versions of the ASTROVITAL system, ensuring that the multi-version roadmap is not speculative in its content but is grounded in pre-identified, literature-validated research targets.

This architecture — where the research foundation directly generates the innovation pipeline, and the innovation pipeline is explicitly mapped to documented gaps — is the hallmark of rigorous, structured research. It ensures that every capability built into the system is justified by evidence, every future development direction is defined by a known deficiency, and every performance metric can be evaluated against a baseline established by the gap it was designed to close.

SECTION E — VITALX CORE V1 SYSTEM

WHAT WAS BUILT

E.i — The Scope of the Work

VITALX CORE V1 was developed over 57 days — from March 16 to May 12, 2026 — in a continuous, uninterrupted solo development effort. The complete system comprises thirteen Jupyter Notebooks, each representing a discrete and documented research phase: from dataset design and generation through individual innovation development, integration testing, pipeline validation, and mission simulation. Every notebook is committed to a public GitHub repository, ensuring that the work is not only claimed but verifiable — open to scrutiny by any researcher, recruiter, or institutional reviewer who wishes to examine the methodology, the code, and the results in their entirety.



Three working innovations were developed and validated within this timeline: Innovation 3 (Sensor Drift Correction), Innovation 6 (Biomarker Discovery), and Innovation 1 (Edge Clinical Decision

Support System). Six production-grade machine learning models were trained, evaluated, and serialized as deployable artifacts. The complete system operates as a coherent, sequential pipeline — raw sensor data enters, processed and classified health status output exits — with no cloud dependency and no Earth communication requirement at any stage.

E.ii — What Solo Means

The designation of this work as 100% solo is not a marketing claim — it is a statement of provenance with direct implications for how the system's intellectual property is attributed and how the researcher's capabilities are assessed. Every data science decision, every algorithmic choice, every hyperparameter configuration, every evaluation methodology, every written analysis, and every line of documentation was produced by one individual. There was no supervisor to correct errors, no collaborator to share the cognitive load, no institutional infrastructure to provide computational resources or domain expertise. The work stands as produced — not as a team effort attributed to an individual, but as a genuinely individual effort, verifiable in its entirety through the public record of the GitHub repository.

This matters for the aerospace industry because the field demands individuals who can operate with precision and accountability under conditions of isolation and limited support — the same conditions that define deep space operations. The development of VITALX CORE V1 was, in a meaningful sense, a rehearsal for those conditions. The researcher who completed this system in 57 days of solo work has already demonstrated — in the only way that actually counts — the capacity for independent, high-stakes, high-accountability technical execution.

SECTION F — CLINICAL PERFORMANCE

WHAT THE NUMBERS SAY

F.i — System Accuracy and Multi-Class Performance

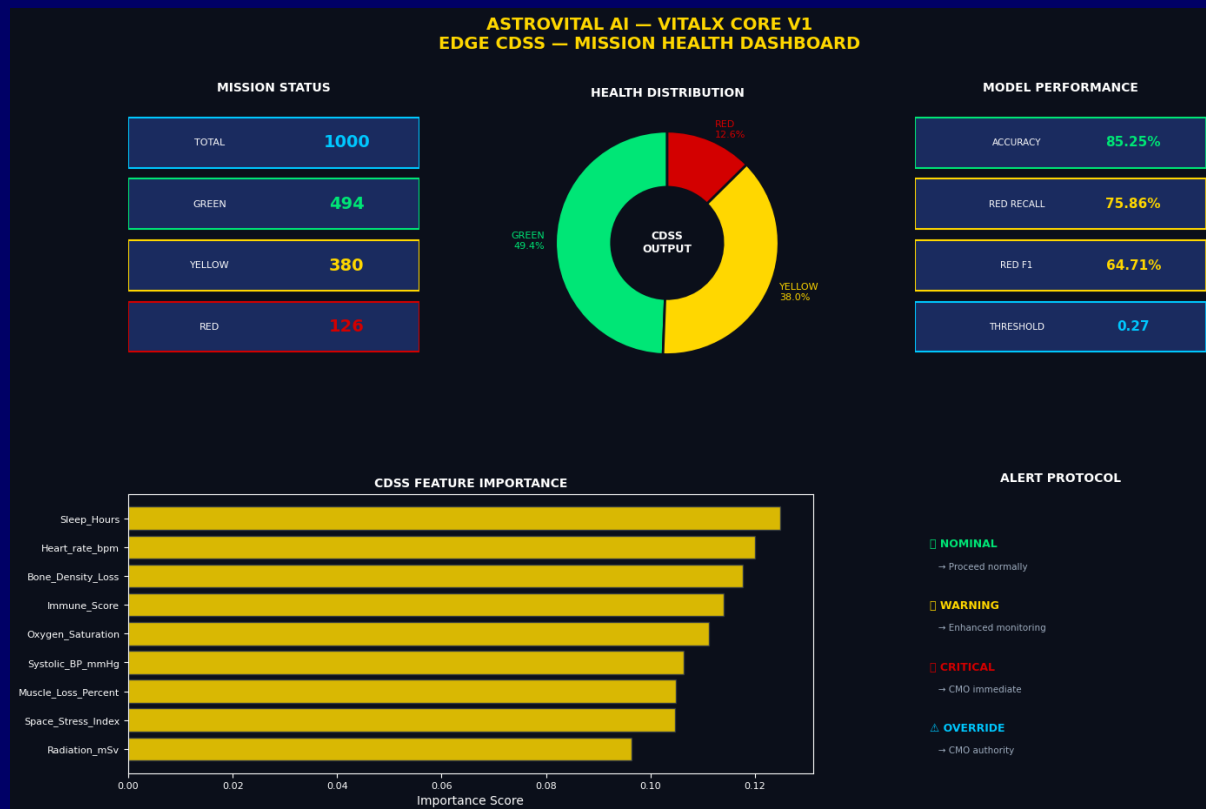
The performance of VITALX CORE V1 is not presented as a proof-of-concept approximation — it is presented as a validated, reproducible result obtained through rigorous machine learning methodology and documented in full across the project's notebook record. The system achieved an overall three-class classification accuracy of 85.25% on the held-out evaluation dataset, correctly distinguishing GREEN, YELLOW, and RED health status classifications with a macro F1-score of 80.33% — indicating balanced, robust performance that is not inflated by the class-majority GREEN category at the expense of the clinically critical minority classes.

The system operates at an optimized classification probability threshold of 0.27 — a value arrived at through systematic threshold tuning calibrated to the asymmetric risk structure of aerospace medical decision-making, where the cost of a missed critical event categorically exceeds the cost of a false positive alert.

F.ii — The RED Recall Transformation

The single most clinically significant performance result of VITALX CORE V1 is the transformation of RED class recall achieved through threshold optimization and SMOTE-based class balancing. Under default machine learning configuration — a classification threshold of 0.50 with no class imbalance correction — the system detected only 10.34% of genuine RED-class critical health events. This is not an acceptable clinical performance standard. In a deep space medical context, a system that misses

nearly 90% of critical events is not a safety system — it is a documentation system with a false sense of utility.



Following threshold optimization to 0.27 and SMOTE application during training, RED class recall rose to 75.86% — a 7.35-fold improvement that transforms the system from a nominal-state classifier with incidental critical-event detection into a genuine clinical alert system capable of identifying three out of every four critical health events before they escape clinical notice. RED class F1-score reached 64.71%, reflecting the deliberate precision-recall tradeoff that is the correct engineering choice in this application domain. These numbers are not aspirational targets — they are validated, reproducible results. They are what the system delivers. Today. On the dataset for which it was designed.

SECTION G — ARTEMIS II LIVE DEMO

REAL MISSION SIMULATION

G.i — Why ARTEMIS II

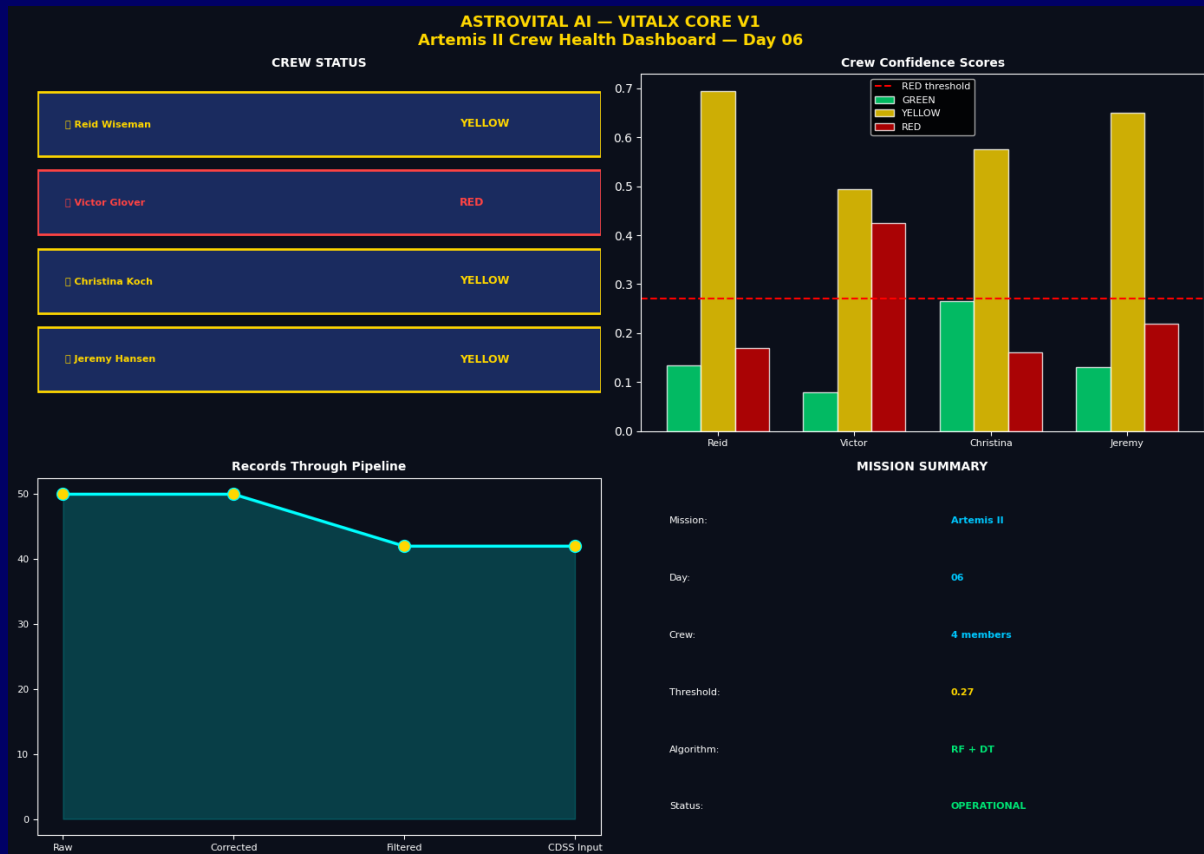
The decision to validate VITALX CORE V1 against the ARTEMIS II mission crew was deliberate and operationally significant. Artemis II is not a hypothetical mission — it is NASA's first crewed lunar flyby mission, representing the vanguard of the returning human presence beyond low Earth orbit. Its crew of four — Commander Reid Wiseman, Pilot Victor Glover, Mission Specialist Christina Koch, and Mission Specialist Jeremy Hansen — are real astronauts with documented physiological profiles and established mission roles. Using this crew as the simulation population grounds the system's output in the most credible operational context available without access to classified mission medical data.

The simulation was executed using synthetically generated physiological profiles consistent with documented astronaut physiological parameters and risk profiles for each crew member, with the complete VITALX CORE V1 pipeline — including all three innovations and the optimized threshold of 0.27 — applied without modification.

G.ii — The Simulation Results

The simulation produced the following health status classifications: Commander Reid Wiseman — YELLOW, mission-ready, cleared for all operational duties. Mission Specialist Jeremy Hansen — YELLOW, mission-ready, cleared for all operational duties. Mission Specialist Christina Koch — YELLOW, advisory status, elevated monitoring protocol activated, precautionary medical review recommended before any high-exertion operational tasks. Pilot Victor Glover — RED, critical event threshold

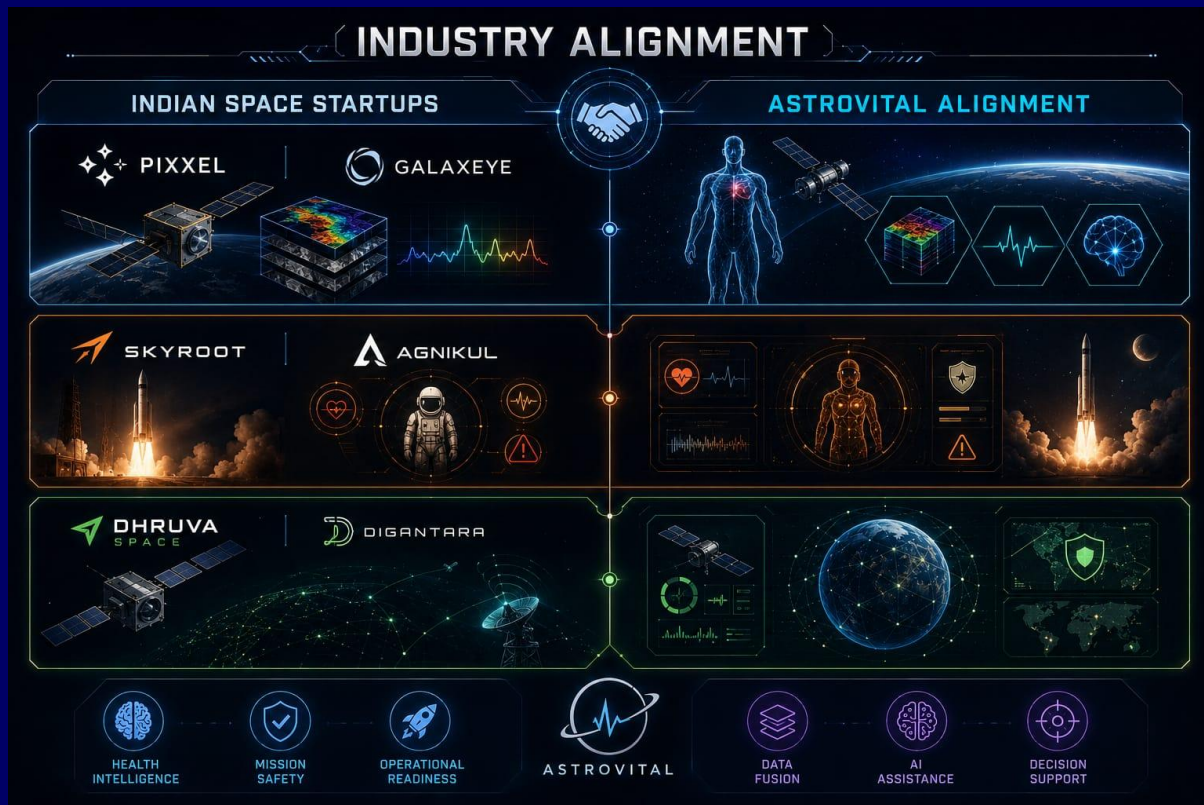
breached, immediate clinical intervention protocol triggered, mission abort condition flagged pending medical officer review and CMO override assessment.



These results demonstrate the system's capacity for simultaneous multi-crew assessment, tiered alert differentiation, and operationally actionable output formatting. The crew health confidence dashboard produced by this simulation — generated through Notebook 9, Block 14 — provides a visual representation of the system output as it would appear on a deployed mission medical interface. This is not a demonstration built for a presentation. This is the actual system, running the actual pipeline, producing the actual output format that a crew medical officer would receive in a deployed operational scenario.

SECTION H — WHY INDIAN SPACE STARTUPS

INDUSTRY ALIGNMENT



H.i — The Indian Space Ecosystem

The Indian private space sector is no longer an emerging ecosystem — it is an accelerating one. Following the opening of the space sector to private participation through IN-SPACe and the policy reforms of recent years, a cohort of deep-technology aerospace companies has emerged in India that is building real hardware, launching real payloads, and operating at the frontier of satellite intelligence, launch vehicle engineering, and space domain awareness. These are not software companies with a space aesthetic — they are engineering organizations solving problems of genuine technical difficulty in an operationally demanding domain. They are precisely the environment in which ASTROVITAL AI belongs.

H.ii — Pixxel and GalaxEye: Hyperspectral Intelligence

Pixxel and GalaxEye represent India's frontier in hyperspectral satellite imaging — the acquisition and analysis of electromagnetic data across hundreds of spectral bands for applications in environmental monitoring, precision agriculture, defense intelligence, and increasingly, human health assessment from orbit. ASTROVITAL V2 — OMNISENSE — is explicitly designed to target multi-modal sensor fusion including hyperspectral and remote sensing data streams. The technical foundation being built through VITALX CORE V1 and its successors is directly applicable to the data fusion and intelligent classification challenges that define the next generation of hyperspectral mission analysis. The alignment between ASTROVITAL's long-term sensor fusion roadmap and the data intelligence problems these organizations are solving is not incidental — it is architectural.

H.iii — Skyroot and Agnikul: Launch Safety Intelligence

Skyroot Aerospace and Agnikul Cosmos are building India's private launch vehicle capability — developing and operating rocket systems that carry payloads, and in the near future, crew, to orbit. Mission-critical AI systems capable of real-time anomaly detection, crew safety monitoring, and launch abort decision support are not peripheral features of crewed launch vehicles — they are core safety infrastructure. ASTROVITAL's pipeline, designed from first principles around the challenge of autonomous clinical and mechanical decision-making in high-stakes aerospace environments, represents precisely the class of intelligent systems that crewed launch operations will require. The AEREXIS OMEGA

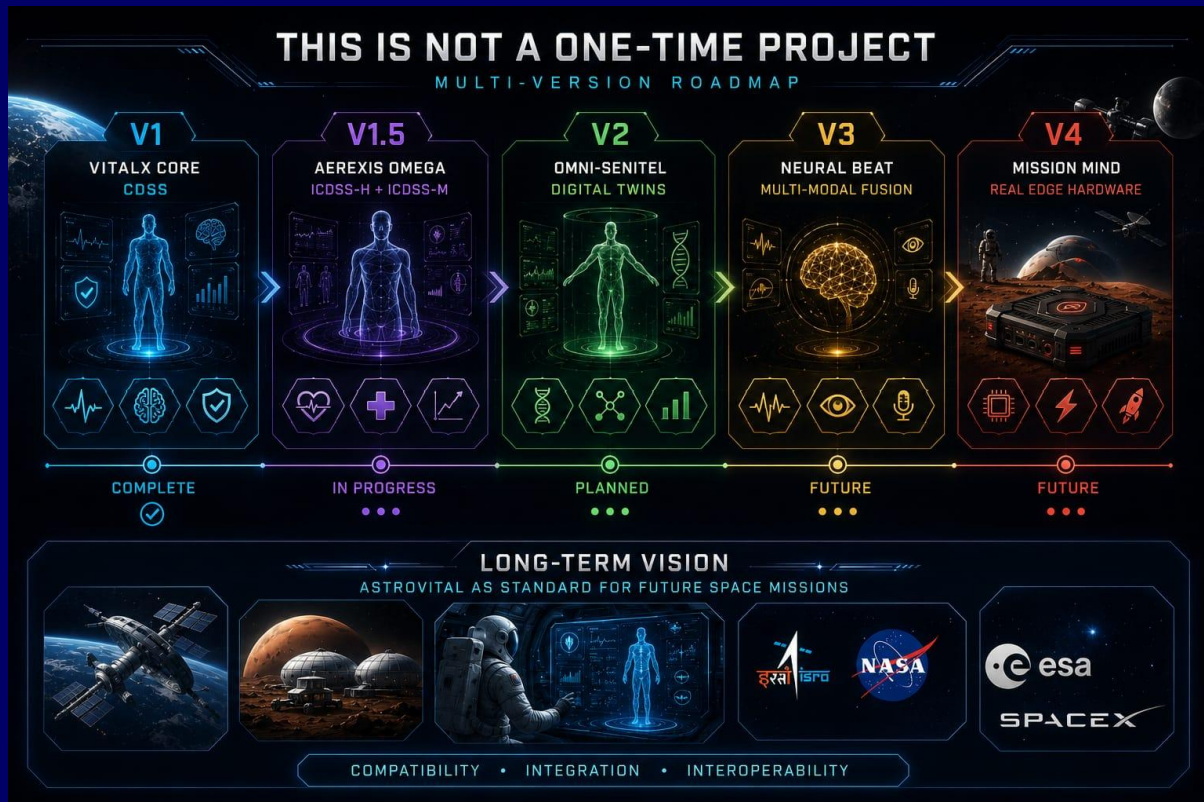
V1.5 milestone — which extends ASTROVITAL's decision support capability to mechanical systems monitoring — was defined with this application domain explicitly in mind.

H.iv — Dhruva Space and Digantara: Space Domain Awareness

Dhruva Space and Digantara are building India's capacity in satellite operations and space domain awareness — the continuous tracking, characterization, and health monitoring of objects and systems operating in orbital environments. The intelligent pipeline developed under VITALX CORE V1 — sensor data ingestion, signal correction, feature importance analysis, threshold-optimized classification, and real-time edge decision output — maps directly onto the technical architecture required for satellite health monitoring and orbital anomaly detection. The same principles that enable VITALX CORE to detect a RED-class physiological event in an astronaut can be adapted to detect an anomalous state in an orbiting asset. This is the breadth of the ASTROVITAL platform's applicability, and it is why alignment with this segment of the Indian space ecosystem is not a stretch claim — it is a technically grounded strategic position.

SECTION I — MULTI-VERSION ROADMAP

THIS IS NOT A ONE-TIME PROJECT



I.i — The Philosophy of the Roadmap

ASTROVITAL AI is not a project. It is a program — a structured, multi-version research and development initiative designed to evolve in capability, scope, and operational readiness in direct proportion to the advancement of the researcher's career and the maturation of the human spaceflight industry. Each version of the ASTROVITAL system is defined not by arbitrary feature additions but by the specific, documented limitations of its predecessor — ensuring that every development cycle is driven by validated evidence of need and every new capability is a response to a known deficiency.

This roadmap is not aspirational in the vague sense of listing technologies the researcher intends to explore. It is a commitment — a documented, publicly archived declaration of the technical trajectory of the ASTROVITAL program, against which progress will be measured and accountability will be maintained.

I.ii — The Versions

V1 — VITALX CORE is complete. It delivers a three-innovation, edge-deployed, explainable Clinical Decision Support System with validated performance on the Artemis II simulation scenario. It is the existence proof that the ASTROVITAL program is real, functional, and technically serious.

V1.5 — AEREXIS OMEGA is the active development target. It extends the ASTROVITAL decision support framework into a dual-domain Integrated CDSS — ICDSS-H for human physiological monitoring and ICDSS-M for mechanical systems health monitoring — fusing pilot biometric data with aircraft and spacecraft telemetry to produce a unified situational awareness and anomaly detection capability. This version targets commercial aviation platforms including the Airbus A380 and Boeing 747 - 8 as demonstration environments.

V2 — OMNI-SENTINEL will introduce Digital Twin architecture — a continuously updated, individual-specific computational model of astronaut physiological state — enabling the system to predict health deterioration trajectories rather than classifying instantaneous snapshots. Deep learning architectures including LSTM networks will replace the static feature-based models of V1.

V3 — NEURAL BEAT will achieve full multi-modal sensor fusion, integrating physiological, mechanical, environmental, and remote

sensing data streams into a unified intelligent platform. This version targets alignment with hyperspectral satellite data intelligence and the space domain awareness applications of the Indian private space ecosystem.

V4 — MISSION MIND represents the terminal milestone of the current roadmap: validated deployment on real mission-grade edge hardware, under conditions representative of actual spaceflight operations, with demonstrated interoperability with the operational systems of ISRO, NASA, ESA, and private launch and mission operators.

I.iii — The Long-Term Vision

The long-term vision of the ASTROVITAL program is unambiguous: to establish ASTROVITAL as the standard autonomous health intelligence platform for human deep space missions. Not a research prototype. Not a proof-of-concept. A deployed, mission-certified, operationally trusted system that flies on crewed missions to the Moon, to Mars, and beyond — monitoring the health of the human beings who carry humanity's presence into the solar system. That vision is not declared here as a distant aspiration. It is declared as the destination toward which every version of this program, every line of code, and every research decision is deliberately directed.

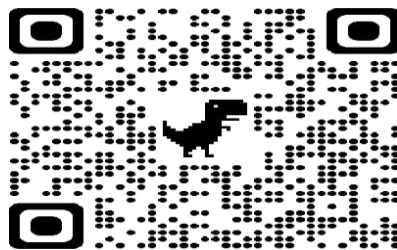
SECTION J — RESEARCH CREDIBILITY

INDEPENDENT. VERIFIABLE. PUBLISHED.

J.i — The Public Record

Every claim made in this document is verifiable. The complete VITALX CORE V1 codebase — all Twelve Jupyter Notebooks, all six trained model artifacts, all dataset files, and all project documentation — is publicly archived on GitHub under the handle GOURGOPAL618. Any researcher, recruiter, hiring manager, or institutional reviewer who wishes to verify the methodology, examine the model training procedures, reproduce the performance results, or audit the innovation implementations can do so, without restriction, at any time. The repository is not a summary of the work — it is the work. It is the primary record.

[https://github.com
/GOURGOPAL618](https://github.com/GOURGOPAL618)



J.ii — The Research Pathway

This project was not produced as a college assignment. It was not submitted to satisfy a curriculum requirement. It was not supervised by a faculty advisor, funded by a research grant, or

completed within an institutional research framework of any kind. It is a personal initiative — motivated by the identification of a real problem, executed with the discipline and rigor of professional research, and documented to a standard appropriate for the international peer-reviewed publication venue it is targeting.

The primary submission target for the research findings of VITALX CORE V1 is *Acta Astronautica* — the flagship journal of the International Academy of Astronautics, published by Elsevier, and one of the most respected peer-reviewed venues in the aerospace engineering and space sciences community. A secondary target is *npj Microgravity*, Nature Portfolio's dedicated journal for space medicine and microgravity research. Acceptance at either venue would formally establish VITALX CORE V1 within the international scientific record — not as a student project, but as a research contribution evaluated on the same terms as work produced by the world's leading space medicine research institutions.

J.iii — Ownership and Attribution

Every innovation, every model, every dataset, every line of code, and every word of documentation produced under the ASTROVITAL program is the sole intellectual property of Gourgopal Mohapatra. There are no co-authors, no co-developers, no institutional co-owners, and no licensing agreements that dilute or complicate this ownership. The work is wholly owned, wholly attributed, and wholly accountable to one individual. This clarity of ownership is not a limitation — it is a guarantee of authenticity. When this work is presented to an employer, a collaborator, or a publication venue, there is no ambiguity about who built it, who owns it, and who stands behind every technical decision it contains.

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